Exploratory Data Analysis (EDA) - House Prices Dataset

1. Dataset Overview:

The dataset contains 1460 rows and 81 columns, including categorical and numerical features. The target variable is 'SalePrice', which represents house prices. Some features have missing values, which were handled accordingly.

2. Missing Values Analysis:

Several columns have high missing values, such as 'Alley' (93.8%), 'PoolQC' (99.5%), and 'Fence' (80.8%). Columns with excessive missing data were dropped, while others were imputed using median/mode values.

3. SalePrice Distribution:

The SalePrice distribution is right-skewed, meaning most houses are on the lower price range with a few extremely expensive ones. A log transformation was applied to normalize the data.

4. Correlation Analysis:

The strongest correlations with SalePrice are:

- OverallQual (0.79): House quality significantly impacts price.
- GrLivArea (0.71): More living space increases price.
- GarageCars (0.64) & GarageArea (0.62): Larger garages lead to higher house prices.
- TotalBsmtSF (0.61) & 1stFlrSF (0.61): Larger basements and first floors also increase value.

5. Key Visualizations:

- Boxplots show that higher OverallQual and newer YearBuilt homes have higher prices.
- The heatmap confirms that some features have little correlation with SalePrice.
- The pairplot highlights linear relationships between top correlated features and SalePrice.

6. Conclusion:

EDA provided valuable insights into the dataset. The most influential features were identified, missing values were handled, and data transformations were applied to improve the analysis. This prepared the dataset for predictive modeling.